

Dissertation / Project / Project Work Title:
Evaluating Graph-Based Retrieval-Augmented Generation:
Comparative Analysis and Quality Improvements over
Standard RAG Systems
Course No.: AIMLCZG628T

Course Title: Dissertation / Project / Project Work

Dissertation / Project /Project Work Done by:

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**Degree Program: MTech. Artificial Intelligence &
Machine Learning**

**Research Area: Conversational AI, Information
Retrieval, and Knowledge Representation**

**Dissertation / Project Work carried out at
WeblinIndia, Ahmedabad**



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1. Broad Area of Work

The broad area of this project lies at the intersection of **Conversational AI**, **Information Retrieval (IR)**, and **Knowledge Representation**. It involves building intelligent question-answering agents that can fetch and utilize external knowledge effectively. Key sub-domains and technologies in this broad area include:

- **Natural Language Processing (NLP):** Understanding user queries and extracting relevant information from text.
- **Information Retrieval:** Searching and retrieving relevant documents or knowledge snippets to ground the AI's responses.
- **Knowledge Representation:** Using structured knowledge (e.g., knowledge graphs) to store facts and relationships for reasoning.
- **Automated Reasoning:** Leveraging both retrieved information and stored knowledge to perform multi-step reasoning and answer complex questions.
- **Machine Learning & LLMs:** Utilizing Large Language Models and fine-tuning techniques to generate natural language answers based on the retrieved context.

2. Background

In the current work environment, there is a strong push to build reliable question answering systems over internal data. The organization maintains a complex and highly interlinked database, with strict accuracy expectations for every response. A standard RAG setup was tried on this data. It struggled to consistently retrieve the right context for edge cases and multi hop queries, and small retrieval errors led to wrong answers that were not acceptable in production.

This project fits into that gap. It looks at going beyond a simple vector-based RAG pipeline and studies whether adding graph style structure on top of the same data can help. The aim is to see if a graph-based RAG setup can better capture relations in the existing database and improve answer quality, especially in areas where precision is critical.

3. Objectives

The objectives of my project are as follows:

- Build a standard RAG based QA system using a vector store and a local LLM.
- Build a graph-based RAG system that uses a knowledge graph on the same data.
- Use a small public QA dataset to benchmark both systems on the same set of questions.
- Study how both systems behave on complex and multi hop queries where accuracy is important.
- Try few shot prompting or light tuning if needed to improve answer quality.
- Compare results and summarize when graph based RAG gives clear gains over standard RAG.

4. Scope of Work

The project builds two systems. One is a standard RAG that uses text retrieval. The other uses a knowledge graph for retrieval. Both will be built using LangChain or LlamaIndex. A small open dataset will be used to test them. The systems will run with local models around 0.5B to 1B in size. Few-shot prompting will be tried if needed. The aim is to see how both systems perform, especially on harder questions that need connecting facts.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Literature review and outline	09 Nov 2025 – 24 Nov 2025	Read work on RAG and graph-based RAG, fix problem statement, freeze tools and dataset, prepare outline.
Standard RAG system development & Graph based RAG system development	25 Nov 2025 – 21 Dec 2025	Design & Development Activity
Testing	22 Dec 2025 – 11 Jan 2026	Software Testing, User Evaluation & Conclusion
Evaluation and comparison	12 Jan 2026 – 24 Jan 2026	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Report writing and finalization	25 Jan 2026 – 01 Feb 2026	Final Review and submission of Dissertation

6. Literature References

The state of the art is the base for any successful research project. In the current project, the literature is inclined towards RAG and graph-based RAG for conversational and question answering systems. The following references are from the preliminary review.

- [1] Ontotext. *What is Graph RAG. Ontotext Fundamentals*, 2023.
<https://www.ontotext.com/knowledgehub/fundamentals/what-is-graph-raa/>
- [2] P. Lewis et al. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020*, arXiv:2005.11401.
<https://arxiv.org/abs/2005.11401>
- [3] Microsoft Research. *Welcome – GraphRAG (project documentation site)*, 2024.
<https://microsoft.github.io/graphrag/>
- [4] J. Liu et al. *How Much Can RAG Help the Reasoning of LLM?* arXiv:2410.02338, 2024.
<https://arxiv.org/html/2410.02338v1>
- [5] A. Gupta. *Fundamental Failure Modes in RAG Systems. PromptQL Blog*, 2025.
<https://promptql.io/blog/fundamental-failure-modes-in-rag-systems>
- [6] H. Han et al. *Retrieval-Augmented Generation with Graphs (GraphRAG). arXiv:2501.00309*, 2024, survey on GraphRAG.
<https://arxiv.org/abs/2501.00309>
- [7] H. Liu, S. Wang, J. Li. *Knowledge Graph Retrieval-Augmented Generation via GNN-Guided Prompting. COLM 2025*.
<https://openreview.net/forum?id=R1NWMEsEj>

7. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Darshan Derasari	Helly Raval
Qualification	Master of Computer Application (MCA)	MCA and PhD in AI Domain
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8. Remarks of the Supervisor

I am pleased to acknowledge the student's initiative on the upcoming project, "Evaluating Graph-Based Retrieval-Augmented Generation: Comparative Analysis and Quality Improvements over Standard RAG Systems." His project proposal demonstrates a solid understanding of advanced AI concepts and a clear plan to explore graph-based methods to enhance retrieval and contextual reasoning in RAG architectures.

He approaches the work with analytical thinking and technical competence, outlining a structured comparison between standard RAG systems and a graph-based alternative. His planned prototype is designed to translate these ideas into a functional model.

Throughout the planning process, he has shown independence, consistency, and a

thoughtful approach to problem-solving. This project is expected to further strengthen his skills in natural language processing, knowledge graph modeling, and system evaluation.

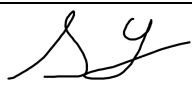
Information about the Supervisor:

Senior Project Manager @ WEBLINE SOLUTIONS PRIVATE LIMITED

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SECOND SEMESTER OF ACADEMIC YEAR 2021-2022

AIMLCZG628T: Dissertation OUTLINE

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Signature of Student	Signature of Supervisor	Signature of Additional Examiner
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